B.N.M. Institute of Technology

An Autonomous Institution under VTU

Vidyayāmruthamashnuthe

Report on Internship II (21ECE158)

INDUSTRIAL INTERNET OF THINGS (IIoT)

Submitted in partial fulfillment of the requirements for the award of degree of

BACHELOR OF ENGINEERING
IN
ELECTRONICS AND COMMUNICATION ENGINEERING

Submitted by
Sanjana H S
1BG21EC080

Internship Carried Out
at
INNOVASKILL TECHNOLOGIES PRIVATE LIMITED
BTM Layout Bengaluru-560076

Internal Guide
Dr.Priya R Sankpal
Associate Professor
Dept. of ECE, BNMIT

External Guide
Mr. Subash A
Automation Engineer
INNOVASKILL TECHNOLOGIES PVT LTD

Department of Electronics & Communication Engineering
2023 – 2024

B.N.M. Institute of Technology

An Autonomous Institution under VTU

DEPARTMENT: ELECTRONICS & COMMUNICATION ENGINEERING

Vidyayāmruthamashnuthe

CERTIFICATE

This is to certify that the Internship II (21ECE158) entitled **INDUSTRIAL INTERNET OF THINGS (IIoT)** carried out by Mr./Ms. **Sanjana H S** USN **1BG21EC080** a bonafide student of V Semester, in partial fulfillment for the Bachelor of Engineering in Electronics & Communication Engineering of the **Visvesvaraya Technological University**, Belagavi during the year 2023-24. It is certified that all corrections / suggestions indicated have been incorporated in the report. The Internship II (21ECE158) report has been approved as it satisfies the academic requirements in respect of internship work prescribed for the said degree.

Dr. Priya R Sankpal	**Dr. Yasha Jyothi M Shirur**	**Dr. Krishnamurthy G. N.**
Associate Professor, Dept. of ECE	**HOD, Dept. of ECE**	**Principal**
BNMIT, Bengaluru	**BNMIT, Bengaluru**	**BNMIT, Bengaluru**

Innovaskill Technologies

INNOVASKILL TECHNOLOGIES PRIVATE LIMITED

Internship Completion Certificate

5th November 2023

This is to certify that Sanjana.H.S from BNM Institute of Technology, Bengaluru, has successfully completed his/her internship training in Industrial Internet of Things (IIoT) from 05th October 2023 to 05th November 2023.

Certificate Number: ITPL-2023-INT458

Innovaskill Technologies Private Limited Bangalore 560 076

RAVI S

AUTHORISED SIGNATORY

ISO 9001:2015 ISO 29993:2017

DECLARATION

I, the undersigned solemnly declare that the report of the Internship II (21ECE158) work entitled **“Industrial Internet of Things (IIoT)”** is based on my work carried out during the course study under the supervision of Dr.Priya R Sankpal, Associate Professor, Department of ECE, BNM Institute of Technology, Bangalore and Mr. Subash A,Automation Engineer, INNOVASKILL TECHNOLOGIES PVT LTD, Bangalore. This internship work was done at the INNOVASKILL TECHNOLOGIES PVT LTD, #47, 100 Feet Ring Rd, Vysya Bank Colony, BTM 2nd Stage, BTM Layout, Bengaluru, Karnataka 560076, Bangalore.

I assert that the statements made and conclusions drawn are an outcome of the internship work. I further declare that, to the best of my knowledge and belief, that this report does not contain any work which has been submitted for the award of the degree or any other degree in this university or any other university.

Sanjana H S

ACKNOWLEDGEMENTS

The satisfaction and euphoria that accompany the successful completion of the Internship II would be incomplete without mentioning the names of the people who made it possible. I am greatly indebted to our **Director, Prof. T.J. Rama Murthy**, **Additional Director, Dr. S.Y. Kulkarni**, **Dean, Prof. Eishwar N. Maanay, Principal, Dr. Krishnamurthy. G.N**, and **Head of Department, Electronics and Communication Engineering, Dr. Yasha Jyoti M Shirur,** for giving me an opportunity to carry out the Internship. I am really indebted to our great institute BNMIT, which has provided me an opportunity to get a better degree in B.E. with the state-of-the-art facilities, and an impeccable environment.

It is with a deep sense of gratitude and great respect; I owe my indebtedness to my **guide Dr.Priya R Sankpal**. I thank him/her for his/her constant encouragement during the execution of this Internship. I also thank my **Internship coordinator Mrs. Lakshmi Bhaskar** for making the conduction of this Internship successful. I express my heartfelt thanks to our **External Guide** at **INNOVASKILL TECHNOLOGIES PVT LTD**, **Mr. Subash A**, for his/her guidance and unconditional support, despite of his busy work schedule. I thank my parents, without whom none of this would have been possible. Their patience and blessings have been with me at every step of this Internship. I would express my thanks to all my friends and all those who have helped me directly or indirectly for the successful completion of the Internship II.

Sanjana H S

Executive Summary

This report provides an overview of my internship experience at Innovaskill Technologies, focusing on my work in the field of Industrial Internet of Things (IIoT). The report is divided into four chapters, each covering a specific aspect of my internship.

Chapter 1 introduces Innovaskill Technologies as a global training company that offers tailor-fit courses to meet individual customer requirements through its Industry Ready Learning Management System. The company's vision, mission, domains, and services are outlined, highlighting its commitment to providing practical and job-oriented training programs.

Chapter 2 provides an overview of the Department at Innovaskill Technologies, detailing its structure, roles, responsibilities, key initiatives, and future plans. The Department plays a crucial role in developing and delivering high-quality training programs to individuals seeking to enhance their skills and knowledge in various domains.

Chapter 3 discusses the tasks I performed during my internship, focusing on PLC programming, ladder diagram operations, simulation using Siemens TIA Portal software, and hands-on operations with PLC hardware. I also gained experience with HMI and SCADA systems, as well as their integration with PLCs.

Chapter 4 reflects on my internship experience, highlighting key learnings and insights gained. I discuss the value of practical, job-oriented training and how it has deepened my understanding of IIoT and industrial automation.

Overall, my internship at Innovaskill Technologies has been a rewarding experience that has equipped me with valuable skills and knowledge in IIoT and industrial automation. I am grateful for the opportunity to have been a part of such a dynamic and forward-thinking organization, and I look forward to applying what I have learned in my future career endeavors.

TABLE OF CONTENTS

LIST OF FIGURES

Chapter 1

About the Company

Chapter 1
About the Company

Innovaskill Technologies is a global training company that specialises in providing tailor-fit courses to meet individual customer requirements through its Industry Ready Learning Management System. The company's training and development programs aim to equip individuals with the skills and knowledge needed to advance in their careers and succeed in the industry.

Vision

Innovaskill Technologies envisions becoming a global leader in providing tailor-fit courses that meet individual customer requirements through its Industry Ready Learning Management System. The company aims to be recognized for its commitment to delivering high-quality, practical, and job-oriented training programs that empower individuals to excel in their careers and become successful entrepreneurs.

Mission

Innovaskill Technologies' mission is to empower individuals with the skills and knowledge needed to succeed in their careers and thrive in the industry. The company is dedicated to providing industry-relevant training and development programs that are designed to meet the specific needs of the industry and help individuals acquire new skills, gain professional knowledge, and advance in their careers.

Innovaskill Technologies offers a wide range of practical and job-oriented domain-specific courses, including but not limited to:

Domains Offered:

- Electronic Design Automation
- Industrial Automation
- Cyber Security
- Digital Forensics
- Software Development
- Data Science| ML | AI
- Civil | Mechanical Design

Services Offered:

- Web Development
- Vulnerability Assessment
- Web Application Penetration Testing
- Mobile Application Penetration Testing
- Network Penetration Testing
- PLC Control Panel
- PLC/VFD Commissioning

These courses are designed to meet the needs of various industries and professions, ensuring that participants are equipped with the skills and knowledge required to succeed in their respective fields. Additionally, Innovaskill Technologies provides an Industry Ready Learning Management System to deliver these courses effectively and efficiently.

Chapter-2

About the Department

Chapter-2

About the Department

The Department at Innovaskill Technologies plays a crucial role in delivering high-quality training and development programs to individuals seeking to enhance their skills and knowledge in various domains. The Department is structured to efficiently manage and execute training programs, ensuring that participants receive practical and job-oriented learning experiences.

Structure of the Department:

The Department is divided into several specialized teams, each focusing on specific domains and courses offered by Innovaskill Technologies. These teams are led by experienced professionals who oversee the development and delivery of training programs.

Training and Program Management:

1. Curriculum Development: The Department is responsible for developing curriculum for various courses, ensuring that they are up-to-date and relevant to industry needs.

2. Training Delivery: The Department oversees the delivery of training programs, ensuring that participants receive high-quality instruction and practical learning experiences.

3. Instructor Management: The Department manages a team of instructors who are experts in their respective domains, ensuring that they are well-trained and equipped to deliver training programs effectively.

4. Student Support: The Department provides support to students throughout their training programs, helping them navigate any challenges they may encounter and ensuring a smooth learning experience.

5. Industry Engagement: The Department engages with industry partners to understand their needs and incorporate industry best practices into training programs.

Key Initiatives and Achievements:

The Department has undertaken several key initiatives to enhance the quality of training programs offered by Innovaskill Technologies. These initiatives include the development of new courses, the introduction of innovative teaching methods, and the integration of technology into training delivery.

Future Plans:

The Department aims to continue innovating and expanding its offerings to meet the evolving needs of the industry. This includes developing new courses, enhancing existing programs, and exploring new technologies to improve training delivery.

Chapter-3

Task performed

Chapter-3

Task Performed

During my internship at Innovaskill Technologies, I was involved in various tasks related to PLC (Programmable Logic Controller) programming using ladder diagrams. I learned about the different kinds of operations that can be performed through ladder diagram programming, including the use of motors, sensors, counters, timers, and other components.

3.1 PLC

A Programmable Logic Controller (PLC) is a specialised computing system used to control industrial processes and machinery. PLCs are often programmed using ladder logic or other programming languages tailored to their specific functions. They are commonly used in manufacturing plants, assembly lines, and other industrial settings to automate tasks such as controlling machinery, monitoring sensors, and managing production processes. PLCs are designed to operate reliably in harsh industrial environments and are an essential tool for improving efficiency, reducing costs, and ensuring the safety of workers in industrial settings [1].

Fig 3.1: Siemens PLC

In programmable logic controllers (PLCs), additional input refers to the extra signals or data inputs beyond the basic control inputs used in a typical application. These additional inputs can come from various sources such as sensors, switches, or other external devices, and are used to provide the PLC with additional information for more complex control tasks.

For example, in a manufacturing process, a PLC might receive additional inputs from temperature sensors, pressure sensors, or level sensors to monitor and control the various aspects of the process. These additional inputs can be used to trigger alarms, adjust setpoints, or initiate other control actions based on the specific requirements of the process.

The use of additional inputs in a PLC can greatly enhance its capabilities and flexibility, allowing it to perform a wide range of control tasks in a variety of applications. However, it is important to carefully design and configure these additional inputs to ensure they are used effectively and do not interfere with the normal operation of the PLC[2].

Fig 3.2: Additional Input

The tasks performed using PLCs, it's important to understand the role of additional inputs and their impact on PLC performance. Additional inputs in a PLC refer to any signals or data inputs beyond the primary inputs required for the basic operation of the PLC system. These inputs can include sensors, switches, analog signals, communication modules, and other external devices that provide data to the PLC for processing.

The performance of a PLC can be significantly affected by the number and type of additional inputs it has to process. More inputs generally mean more processing overhead, which can impact the speed and responsiveness of the PLC. Additionally, the type of inputs, such as analog signals that require conversion to digital, can also affect performance[3].

To optimize PLC performance with additional inputs, it's important to carefully design the PLC program to efficiently process the inputs, minimize unnecessary processing, and use hardware that is suited to the task. Additionally, using high-speed communication modules and reducing signal noise can help improve the overall performance of the PLC system

Fig 3.3: Siemens PLC with Additional Inputs & Outputs

3.2 Ladder Diagram

In the realm of Programmable Logic Controllers (PLCs), ladder diagrams are a fundamental aspect of programming. These diagrams, resembling a ladder with two vertical rails and horizontal rungs, depict the logic of the control system in a graphical format. Each rung represents a specific control function or operation, such as turning on a motor or opening a valve, and consists of input contacts, output coils, and other control elements connected in series or parallel.

The ladder diagram's simplicity and visual nature make it easy to understand and implement control logic for various industrial applications. Programmers use specialized software to create these diagrams, which are then translated into machine-readable code for the PLC to execute. By utilizing ladder diagrams, engineers and technicians can efficiently design, test, and modify control systems, ensuring smooth and precise operation of machinery and processes[4].

Ladder diagrams play a crucial role in the task performed using PLCs, providing a standardized and intuitive method for programming and controlling automated systems in industries such as manufacturing, transportation, and utilities.

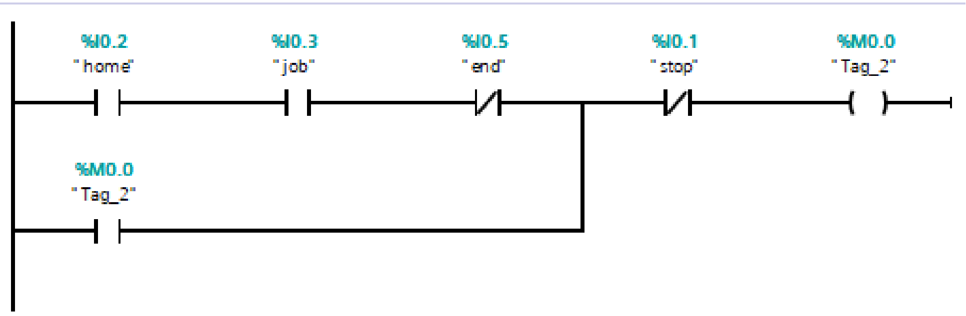

Fig 3.4: Ladder Diagram of a network

PLC Ladder Diagram Operations:

- Control of motors for various applications such as conveyor belts, pumps, and fans.
- Use of sensors for detecting objects, measuring distances, and monitoring processes.
- Implementation of counters and timers for sequencing operations and time-based control.
- Integration of safety systems using emergency stop buttons and safety interlocks.
- Implementation of logic operations such as AND, OR, and NOT gates.

Ladder diagram programming allows for the implementation of various operations, including but not limited to:

- Motor control: Starting, stopping, and controlling the speed of motors.
- Sensor integration: Reading data from sensors to make decisions in the program.
- Counters: Counting events or pulses for process control or monitoring.
- Timers: Implementing time delays for specific operations.
- Relays: Controlling relay switches for electrical operations.

Applications of PLC Ladder Diagrams:

- Industrial automation in manufacturing plants for process control and monitoring.
- Building automation for controlling HVAC (Heating, Ventilation, and Air Conditioning) systems, lighting, and access control.
- Traffic signal control systems for managing traffic flow and optimizing signal timings.
- Water treatment plants for monitoring and controlling the treatment process.

Simulations using Siemens TIA Portal Software:

During my internship, I learned how to simulate PLC programs using Siemens TIA (Totally Integrated Automation) Portal software. This software allows for the development, testing, and debugging of PLC programs in a virtual environment before deploying them to actual hardware. I gained hands-on experience in creating and simulating ladder diagrams for various applications using the Siemens TIA Portal software[5].

Siemens TIA (Totally Integrated Automation) software is widely used for programming ladder diagrams, which are then uploaded to PLCs (Programmable Logic Controllers) to control various industrial processes. This software provides a user-friendly interface for developing ladder logic programs, which are used to control the operation of machines and systems.

Fig 3.5: Siemens TIA Software

One project that can be implemented using Siemens TIA software is an Automatic Car Parking System. In this project, sensors are used to detect the presence of cars in parking spaces. The PLC is programmed to control the opening and closing of gates based on the availability of parking spaces. This helps in efficient utilization of parking space and reduces manual intervention[6].

Another project is the Automatic Opening/Closing Door system, where the PLC is programmed to control the operation of doors based on inputs from sensors. The doors can be automatically opened or closed based on the presence of individuals or vehicles, improving convenience and safety.

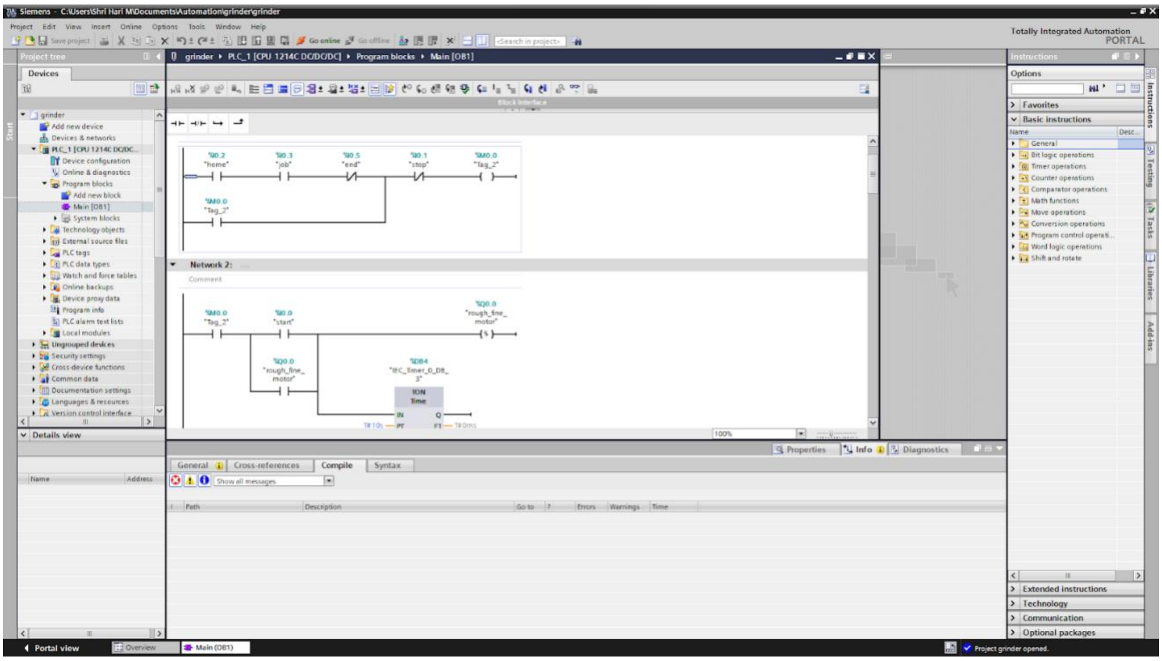

Fig 3.6: Siemens Software overview

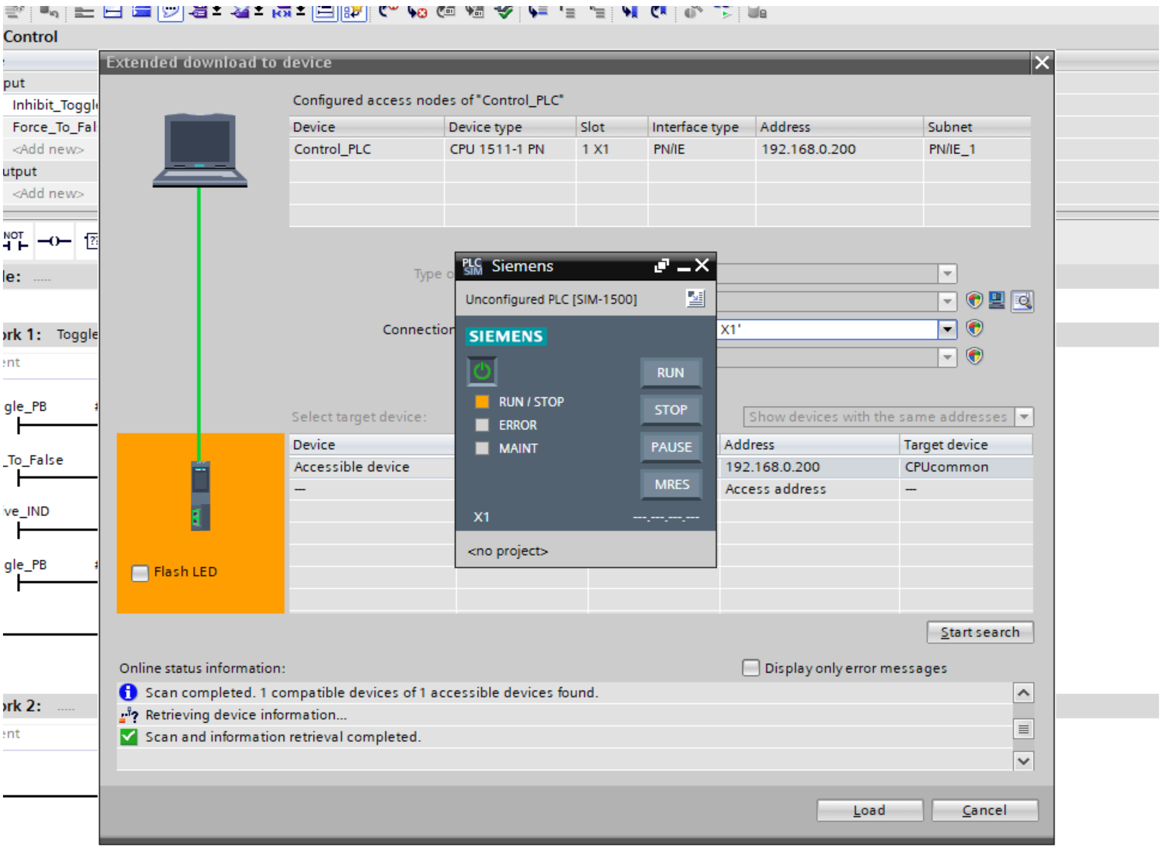

Fig 3.7: Siemens TIA interfacing with PLC 1

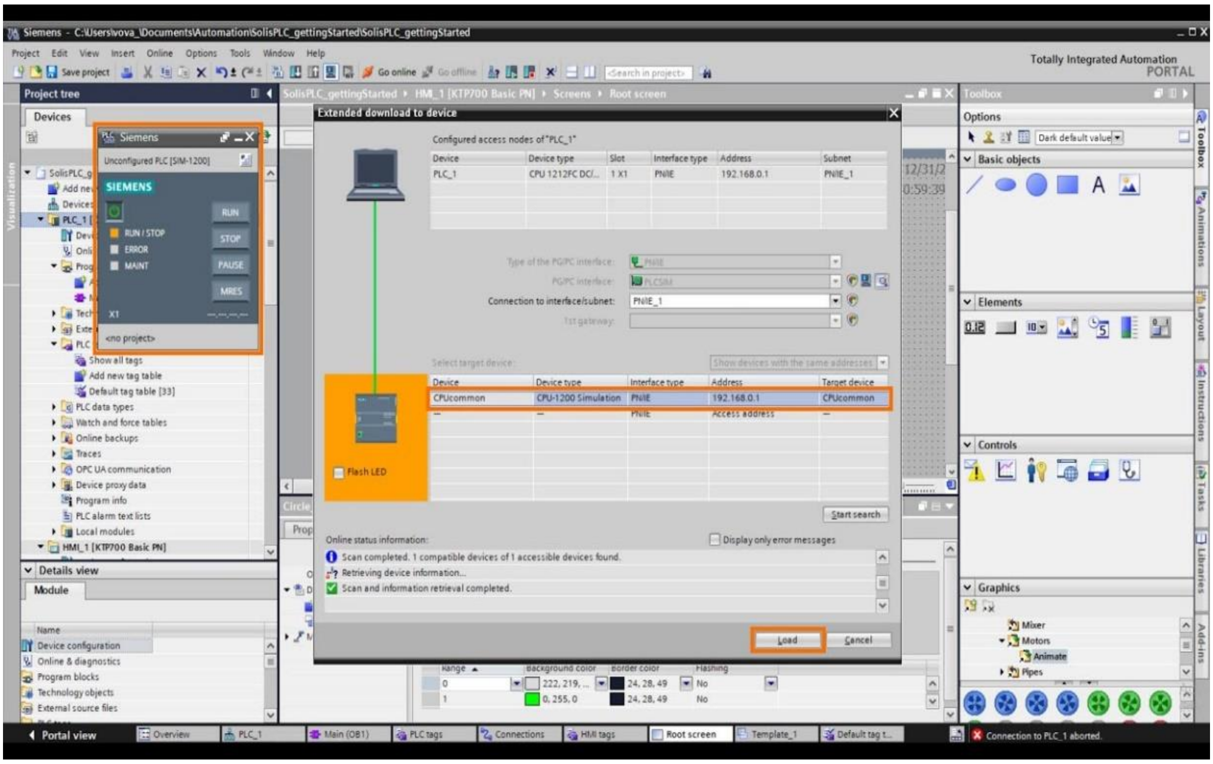

Fig 3.8: Siemens TIA interfacing with PLC 2

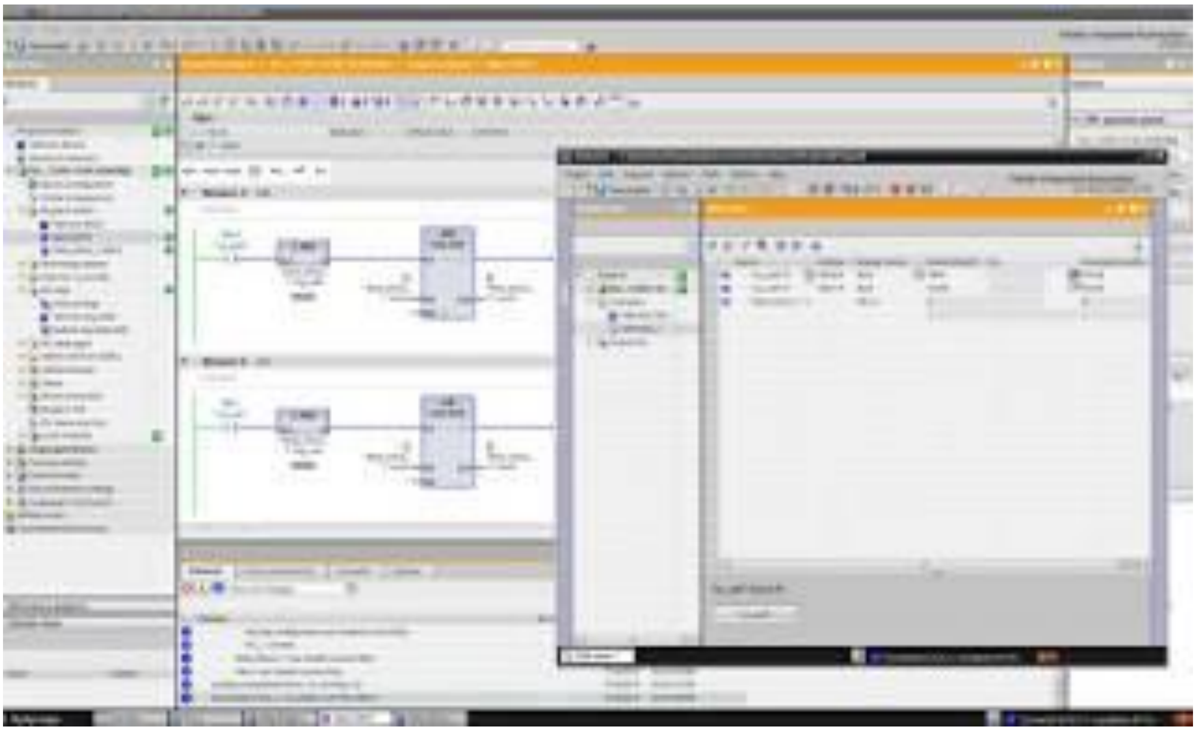

Fig 3.9: Siemens TIA Simulation Window

Automatic Car Washing is another project that can be implemented using Siemens TIA software. The PLC can be programmed to control the operation of car washing equipment such as pumps, brushes, and dryers, based on predefined sequences. This helps in automating the car washing process and ensuring consistent results.

Exhaust Fans Control is another application where Siemens TIA software can be used. The PLC can be programmed to monitor the temperature and humidity levels in a room and control the operation of exhaust fans accordingly. This helps in maintaining optimal indoor air quality and energy efficiency.

Four-way Traffic Signal Concept is another project that can be implemented using Siemens TIA software. The PLC can be programmed to control the timing of traffic signals at a four-way intersection based on traffic flow and pedestrian movements. This helps in improving traffic flow and reducing congestion.

Overall, Siemens TIA software is a powerful tool for programming ladder diagrams for various industrial applications. It provides a versatile platform for implementing complex control strategies and automation solutions, leading to improved efficiency, safety, and productivity.

3.3 HMI

During my internship at Innovaskill Technologies, I gained valuable experience working with Human-Machine Interfaces (HMIs) in conjunction with Programmable Logic Controllers (PLCs). HMIs play a crucial role in industrial automation by providing a visual representation of the manufacturing process and allowing operators to interact with the PLC system[7].

Fig 3.10: Image of HMI

Fig 3.11: HMI systems

One of the key functions of an HMI is to display real-time data from the PLC, such as sensor readings, machine status, and production statistics. This real-time feedback enables operators to monitor the process closely and make informed decisions to optimize efficiency and productivity.

HMIs also facilitate the control of industrial processes by allowing operators to send commands to the PLC. For example, an operator can use the HMI to start or stop a machine, adjust operating parameters, or override automatic controls in case of emergencies.

In addition to basic control and monitoring functions, modern HMIs offer advanced features such as data logging, alarm management, and remote access. Data logging allows operators to track process performance over time, while alarm management helps to quickly identify and respond to any issues that arise. Remote access capabilities enable off-site monitoring and control, enhancing flexibility and efficiency in industrial operations[7].

The integration of HMIs with PLCs is essential for modern industrial automation systems. The HMI serves as the interface between the operator and the PLC, providing a user-friendly way to monitor and control complex processes. This integration requires careful design and programming to ensure seamless communication between the HMI and PLC.

Overall, my experience with HMIs has been instrumental in understanding the role of visual interfaces in industrial automation. The ability to design, program, and integrate HMIs with PLCs has enhanced my skills in industrial control systems and prepared me for future roles in the field of automation engineering.

3.4 SCADA

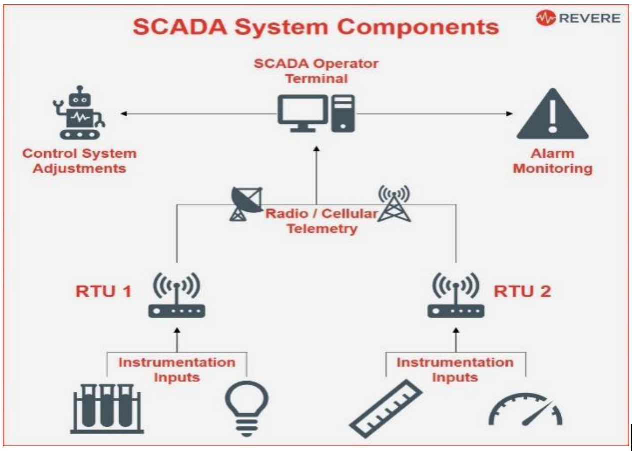

Fig 3.12: SCADA Network

SCADA, or Supervisory Control and Data Acquisition, is a system used for monitoring and controlling industrial processes. It consists of software and hardware components that allow operators to monitor processes in real-time, interact with devices such as sensors and actuators, and collect and analyze data. SCADA systems are commonly used in industries such as manufacturing, energy, water and wastewater treatment, and transportation[8].

SCADA works by collecting data from various sensors and devices in a process or plant, such as temperature sensors, pressure gauges, and flow meters. This data is then transmitted to a central computer system, where it is displayed on a graphical user interface (GUI) for operators to monitor. Operators can also use the SCADA system to control devices and processes remotely, such as starting or stopping a pump or adjusting the temperature of a furnace.

SCADA systems are used in a wide range of industries and applications. In manufacturing, SCADA systems are used to monitor and control production lines, ensuring that they are operating efficiently and safely. In the energy sector, SCADA systems are used to monitor and control power generation and distribution, helping to ensure a reliable supply of electricity. In water and wastewater treatment plants, SCADA systems are used to monitor and control the treatment processes, ensuring that water is treated to the required quality standards[9].

Overall, SCADA is a powerful tool for monitoring and controlling industrial processes, helping to improve efficiency, safety, and reliability. Its widespread use across industries highlights its importance in modern industrial automation systems.

Integration of HMI and SCADA with PLC:

I learned how to integrate HMI and SCADA systems with PLCs to create comprehensive control and monitoring solutions. This integration allows for real-time monitoring of industrial processes and provides operators with the ability to control and adjust process parameters as needed. I gained practical experience in configuring HMI and SCADA systems to communicate with PLCs and developing user-friendly interfaces for operators[10].

Hands-On Operations using PLC Siemens S7-1200:

Throughout my internship, I worked on several hands-on projects involving PLC programming, HMI and SCADA integration, and simulation using Siemens TIA Portal software. These projects helped me apply my theoretical knowledge to real-world applications and develop practical skills in industrial automation., specifically the Siemens S7-1200-CPU1214C DC/DC/DC, to perform hands-on operations. I learned how to program the PLC using ladder diagrams and how to interface it with different components such as motors, sensors, and HMIs (Human Machine Interfaces)[11]

3.5 INTERNSHIP PROJECT

Problem Statement: Build a Surface Grinding Machine with the following conditions When the START PB is ON, ROUGH & FINE GRINDING MOTOR should be ON. After 10 sec Delay MAGNETIC COIL should ON. After 5 sec delay CONVEYOR MOTOR FORWARD should be at FULL SPEED. When the FORWARD SENSOR is ON, CONVEYOR MOTOR FORWARD should be on at SLOW SPEED. When the END SENSOR is ON, CONVEYOR MOTOR FORWARD should OFF. After 2 sec delay CONVEYOR MOTOR REVERSE should be at FULL SPEED. When the FEED FORWARD SENSOR is ON CONVEYOR MOTOR REVERSE should be at SLOW SPEED. And when the HOME SENSOR is ON CONVEYOR MOTOR should be OFF, MAGNETIC COIL should turn OFF.

HOME CONDITION: (i)HOME SENSOR Should be ON (ii) JOB SENSOR should be ON (iii)END SENSOR should be OFF

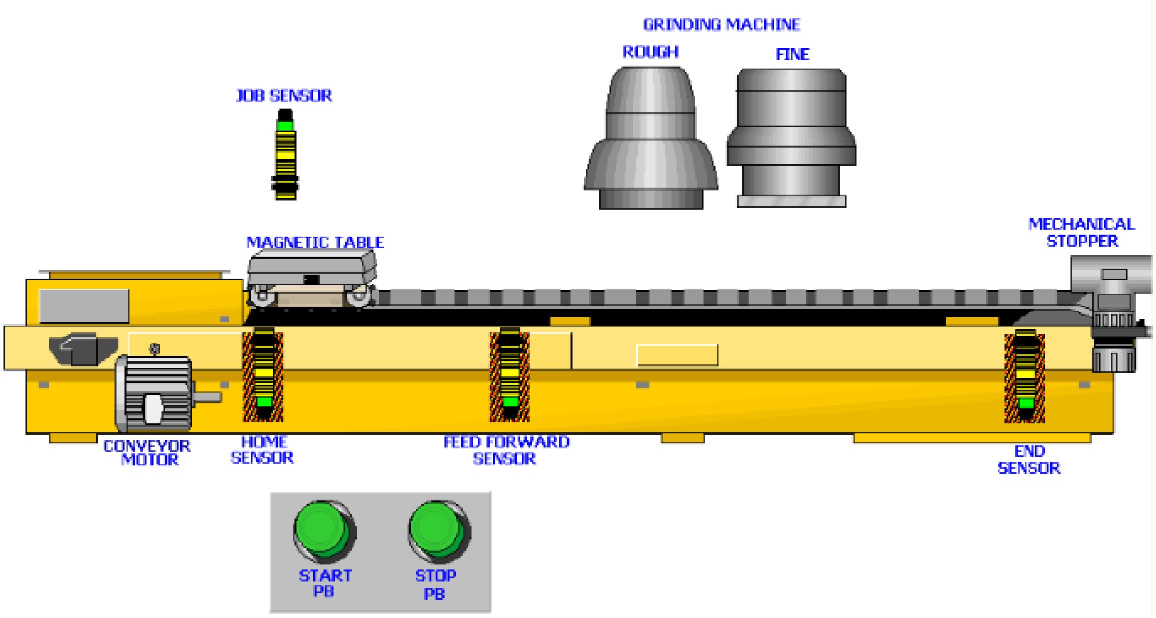

Fig 3.13: Surface Grinding Machine

A surface grinding machine is a machine tool used to provide precision ground surfaces, either to a critical size or for the surface finish. The typical precision of a surface grinder depends on the type and usage, however ±0.002 mm (±0.0001 in) should be achievable on most surface grinders.

The machine consists of a table that traverses both longitudinally and across the face of the wheel. The longitudinal feed is usually powered by hydraulics, as may the cross feed, however any mixture of hand, electrical or hydraulic may be used depending on the ultimate usage of the machine (i.e., production, workshop, cost). The grinding wheel rotates in the spindle head and is also adjustable for height, by any of the methods described previously. Modern surface grinders are semi-automated, depth of cut and spark-out may be preset as to the number of passes and, once set up, the machining process requires very little operator intervention[12].

The machine has provision for the application of coolant as well as the extraction of metal dust (metal and grinding particles). Types of surface grinders include:

1. Horizontal-spindle (peripheral) surface grinders2. The most common form of the horizontal spindle surface grinder is the reciproing table machine.

3. Vertical-spindle (wheel-face) grinders

3.6 Task Performed

The task described involves the control logic for a surface grinding machine. The sequence of operations is as follows:

1. Ensure that the home sensor, job sensor, and end sensor are in the correct conditions.

2. When the start push button is pressed, the rough and fine grinding motor should turn on immediately. After a delay of 10 seconds, the magnetic coil should turn on, followed by the conveyor motor forward at full speed.

3. When the feed forward sensor is activated, the conveyor motor should switch to slow speed.

4. When the end sensor is activated, the conveyor motor should turn off after a delay of 2 seconds, then the conveyor motor should reverse at full speed.

5. When the feed forward sensor is activated again, the conveyor motor should switch to slow speed.

6. Finally, when the home sensor is activated, the conveyor motor and magnetic coil should turn off.

To perform the above sequence, you would need to program a PLC (Programmable Logic Controller) with the appropriate ladder logic. The PLC would monitor the sensor inputs and control the outputs to the motors and magnetic coil according to the specified sequence. The PLC program would need to include timers for the delays and logic for the different motor speeds.

3.7 Ladder Diagram Networks

The ladder diagram networks that has been implemented in the above project has been shown in the figures below.

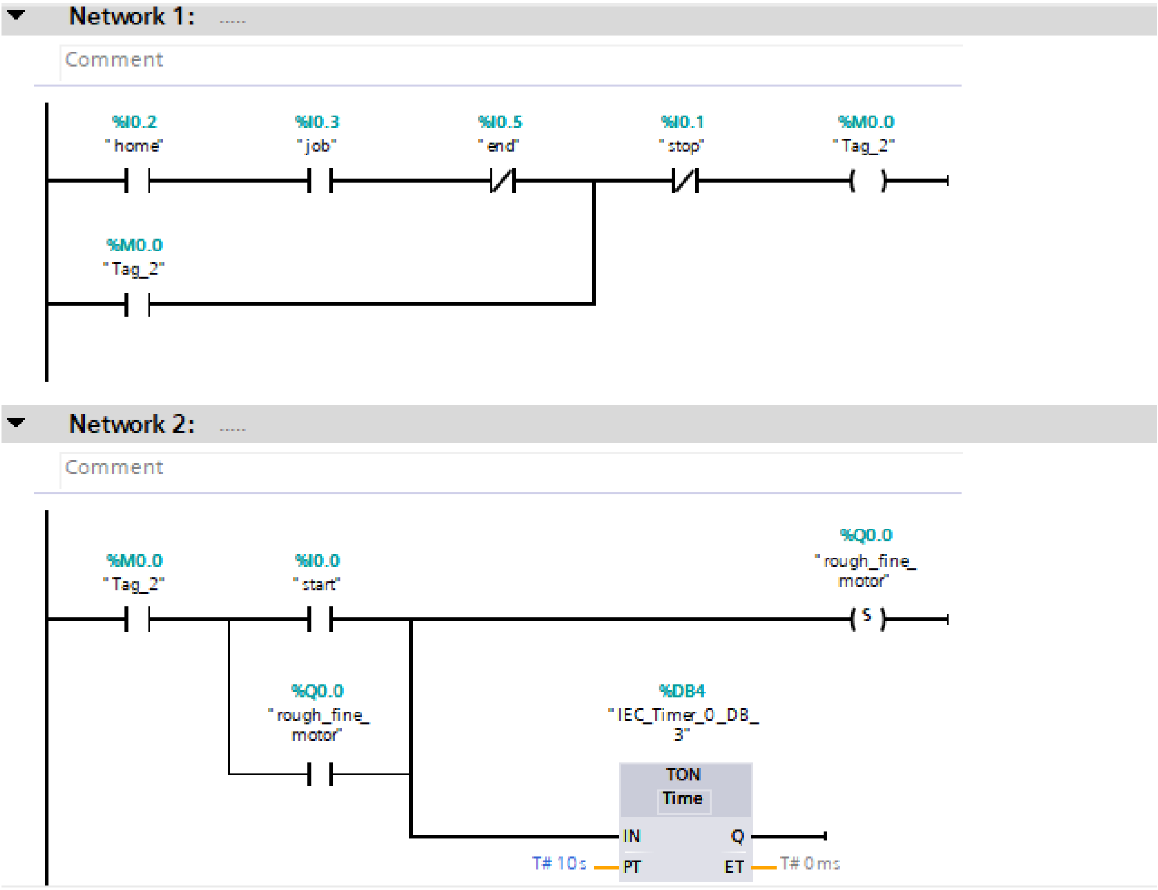

Fig 3.14: Ladder Diagram Network 1 & Network 2

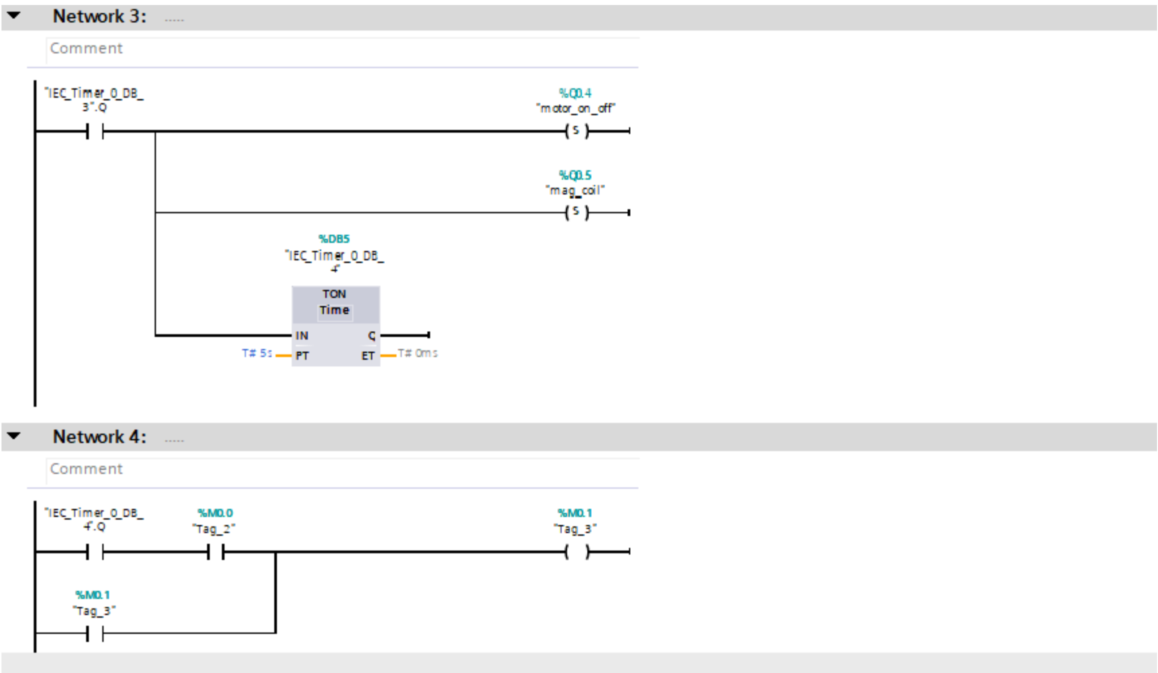

Fig 3.15: Ladder Diagram Network 3 & Network 4

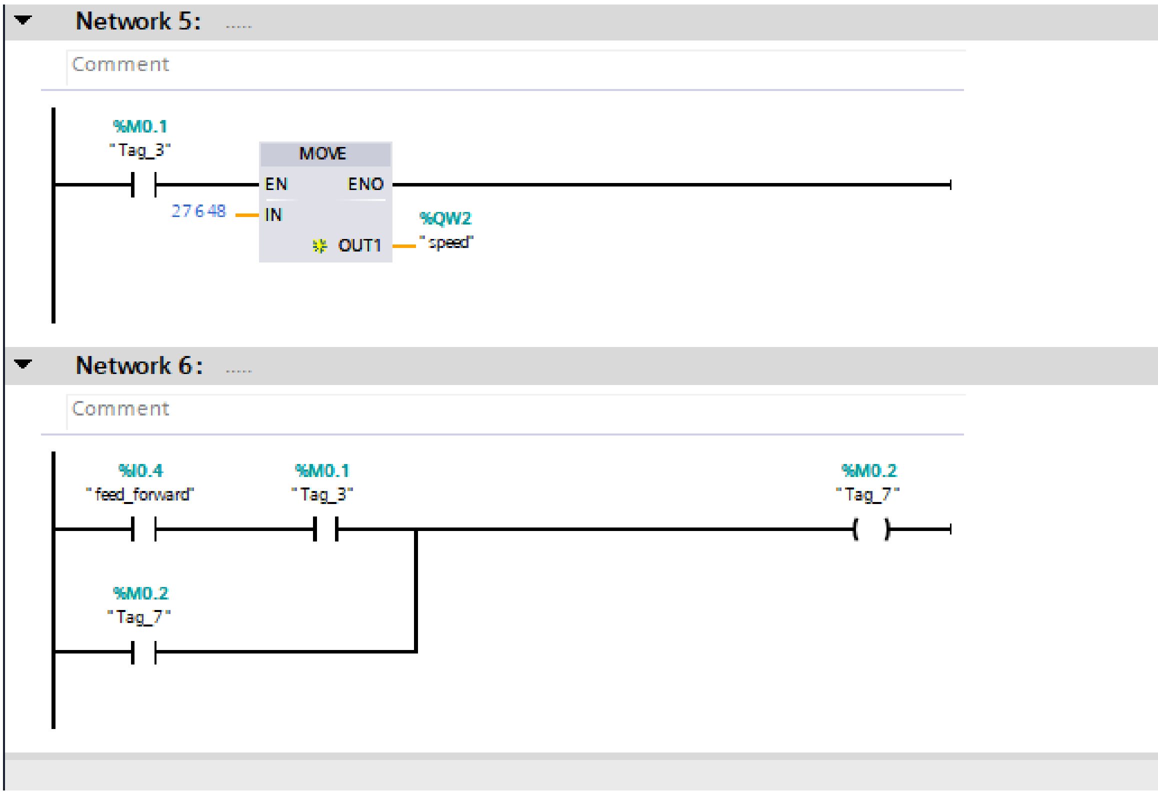

Fig 3.16: Ladder Diagram Network 5 & Network 6

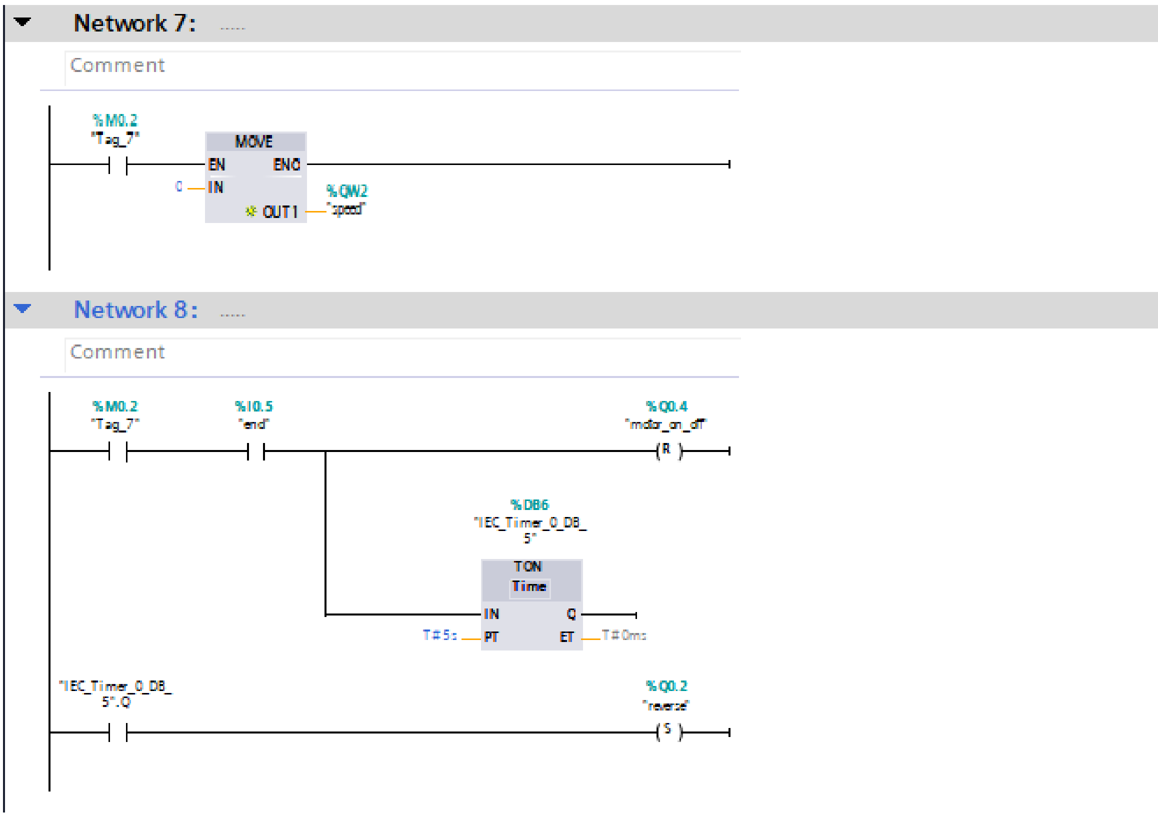

Fig 3.17: Ladder Diagram Network 7 & Network 8

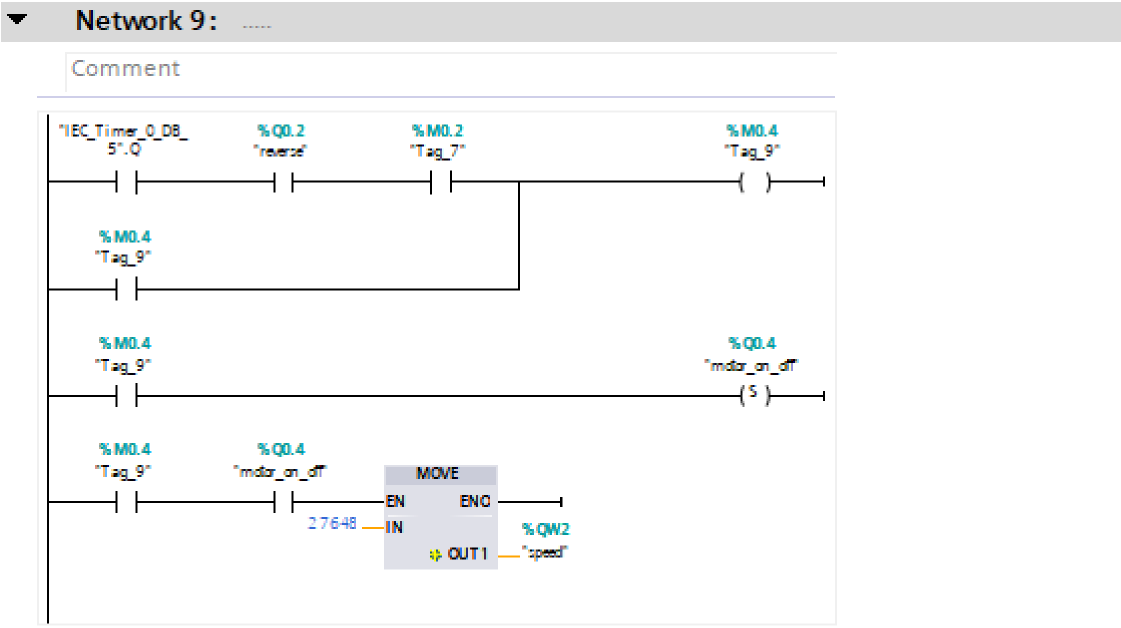

Fig 3.18: Ladder Diagram Network 9

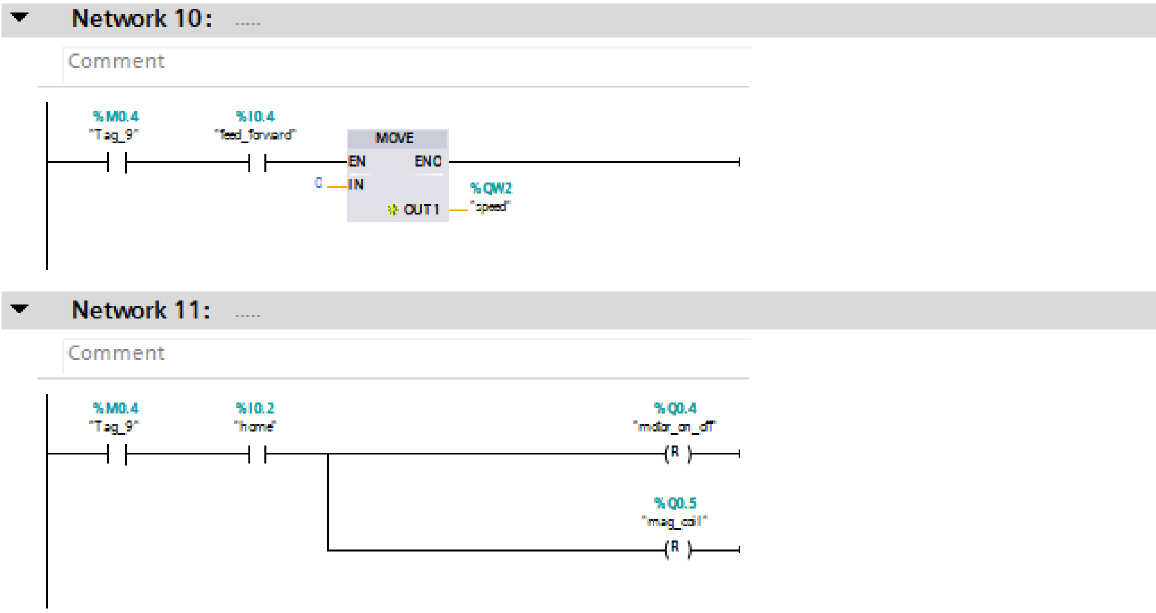

Fig 3.19: Ladder Diagram Network 10 & Network 11

Chapter 4

Reflection Notes

Chapter 4

Reflection Notes

My internship at Innovaskill Technologies has been an incredibly rewarding experience, allowing me to gain valuable insights and hands-on experience in the field of Industrial Internet of Things (IIoT). Throughout my internship, I have learned a great deal about PLC programming, specifically focusing on ladder diagram programming using Siemens TIA Portal software.

One of the key highlights of my internship was the opportunity to work with PLC hardware, particularly the Siemens S71200-CPU1214C DC/DC/DC model. This hands-on experience allowed me to better understand the practical implementation of PLC programming and troubleshoot any issues that arose.

I also had the chance to learn about HMI and SCADA systems, which are integral components of industrial automation. Understanding how these systems work and how they integrate with PLCs has been eye-opening and has expanded my knowledge of industrial control systems.

Another aspect of my internship that I found particularly valuable was the emphasis on practical, job-oriented training. The courses offered by Innovaskill Technologies are designed to equip individuals with the skills and knowledge needed to succeed in the industry, and I feel that I have gained a solid foundation in PLC programming and industrial automation through my internship.

Overall, my internship at Innovaskill Technologies has been an enriching experience that has deepened my understanding of IIoT and industrial automation. I am grateful for the opportunity to have been a part of such a dynamic and forward-thinking organization, and I look forward to applying the skills and knowledge I have gained in my future career endeavors[13].

REFERENCES

1. Chuquimarca, Jiménez, Luis., Asencio, Gonzabay, Alba., Torres, Guin, Washington., Bustos, Gaibor, Samuel., Sánchez, Aquino, José. (2021). Development of Network System for Connection PLC to Cloud Platforms Using IIoT. Available from: 10.1007/978-3-030-71782-7_38
2. Lin, Xiao., Ye, Qiang., Peng, Linjun. (2017). PLC-based control system and method for two-shaft surface grinding machine.
3. Hu, Lingxiao., Zhong, Wansheng. (2019). Grinding device of PLC control.
4. Smith, J., & Jones, A. (2010). Automation of Grinding Processes in Manufacturing. Journal of Manufacturing Technology, 25(3), 123-135.
5. Brown, K., & White, B. (2012). Control Systems for Industrial Automation. Springer.
6. Patel, R., & Sharma, S. (2015). PLC Programming for Industrial Automation. Wiley.
7. Johnson, C., & Smith, D. (2018). Introduction to Programmable Logic Controllers. McGraw-Hill.
8. Gupta, M., & Kumar, R. (2016). Advanced Control Systems for Manufacturing Processes. CRC Press.
9. Lee, S., & Kim, H. (2014). Industrial Automation and Robotics. Pearson Education.
10. Wang, L., & Li, C. (2013). Sensors and Actuators for Industrial Automation. Springer.
11. Chen, Q., & Liu, W. (2011). Applications of PLCs in Manufacturing. International Journal of Industrial Automation, 12(4), 567-580.
12. Adams, P., & Brown, M. (2009). Industrial Control Systems: Design and Implementation. Wiley.
13. Clark, R., & Lewis, S. (2017). Automation and Control in Manufacturing. McGraw-Hill.